

Experimental investigation on hydrogen transfer in coupled metal hydride reactors for multistage hydrogen purification application

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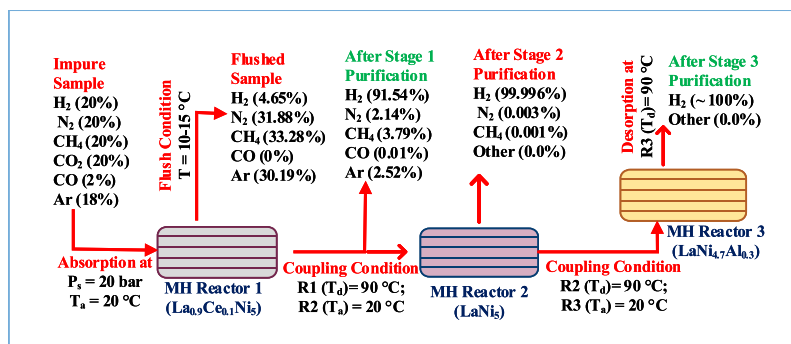
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HIGHLIGHTS

- Demonstrated the purification of hydrogen from gas samples with 80% impurity.
- Presented the hydrogen transfer characteristics between coupled reactors.
- Reported the desired operating conditions for multistage hydrogen purification.
- Performed TCD analysis of gas samples at various stages of purification process.
- Studied poisoning and regeneration characteristics of metal hydride reactors.

GRAPHICAL ABSTRACT



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ABSTRACT

Three stage metal hydride hydrogen separation system was tested, wherein each stage was configured with 1.2 kg of $\text{LaNi}_{4.7}\text{Al}_{0.3}$, LaNi_5 and $\text{La}_{0.9}\text{Ce}_{0.1}\text{Ni}_5$, placed in descending order of their equilibrium bed pressure respectively as R1 ($\text{La}_{0.9}\text{Ce}_{0.1}\text{Ni}_5$) > R2 (LaNi_5) > R3 ($\text{LaNi}_{4.7}\text{Al}_{0.3}$). Hydrogen absorption capacity obtained for R1, R2 and R3 were 15.91 g, 15.93 g and 16.02 g respectively for 3rd activation cycle, performed at 30 bar and 25 °C. For coupled reactor operation, optimum hydrogen transfer was for desorbing reactor at 90 °C and absorbing reactor at 20 °C, which was 12.31 g for R1 to R2 in 350 s, 13.33 g for R2 to R3 in 300 s and 13.38 g for R1 to R2 in 280 s, constituting to 1.03, 1.11 and 1.12 wt% of hydrogen storage respectively. Further, the system was tested for hydrogen purification, wherein inlet sample with 80% impurity (18% Ar, 20% CO_2 , 20% N_2 , 20% CH_4 and 2% CO) and 20% H_2 was used for purification study. According to Thermal Conductivity Detector (TCD) results, the system delivered 91.54%, 99.996% and 99.9999% pure hydrogen respectively for stage 1, stage 2 and stage 3 purification. The system works within low temperature range of 20–90 °C.

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Nomenclature

Symbols

Bed Avg	Average bed temperature, °C
HTF _{in}	Water inlet temperature, °C
HTF _{out}	Water outlet temperature, °C
P _s	Hydrogen supply pressure, bar
R1	Reactor with La _{0.9} Ce _{0.1} Ni ₅
R2	Reactor with LaNi ₅
R3	Reactor with LaNi _{4.7} Al _{0.3}
T _a	Absorption temperature, °C
T _d	Desorption temperature, °C

Abbreviations

ECT	Embedded Cooling Tube
GC	Gas Chromatography
HTF	Heat Transfer Fluid
MH	Metal Hydride
MHPS	Metal Hydride Hydrogen Purification System
PCT	Pressure Concentration Isotherm
PV	Photo Voltaic
TCD	Thermal Conductivity Detector

1. Introduction

Observing consistent rise in global warming, due to dependency on fossil fuels to meet global energy demand, the world is looking for a parallel shift towards the renewable energy utilization technologies to meet energy demand without affecting environment [1–3]. Among different renewable energy sources, hydrogen stands firmly due to its high energy density, and capabilities to replace conventional energy sources effectively [4,5]. As an excellent energy carrier and environmental friendly in nature, H₂ has become perfect option for limiting the rapid increase in CO₂ concentration in the atmosphere and restricting global temperature rise to 1.5 °C. Therefore, the world is gearing towards hydrogen energy economy for fulfilling the various energy demand [6,7]. Hydrogen can be extracted through electrolysis of water, PV electrolysis, decomposition of fossil fuels, thermolysis of water, reforming, biomass conversion, gasification, dark fermentation, bio photolysis, etc. [8–11]. And, it can be utilised for different engineering/industrial applications such as hydrogen fuel cell, food and chemical industries, automobile industries, ceramic and steel industries, weather balloon, synthesis of chemical compounds, space sector, etc. [12–16].

However, most of the engineering application requires pure grade hydrogen for its efficient operation. One of such examples is the use of hydrogen in electric generator coil cooling, wherein the performance of generator in the thermal power plants is directly influenced by the purity level of hydrogen [17,18]. Similarly, efficiency of hydrogen fuel cell

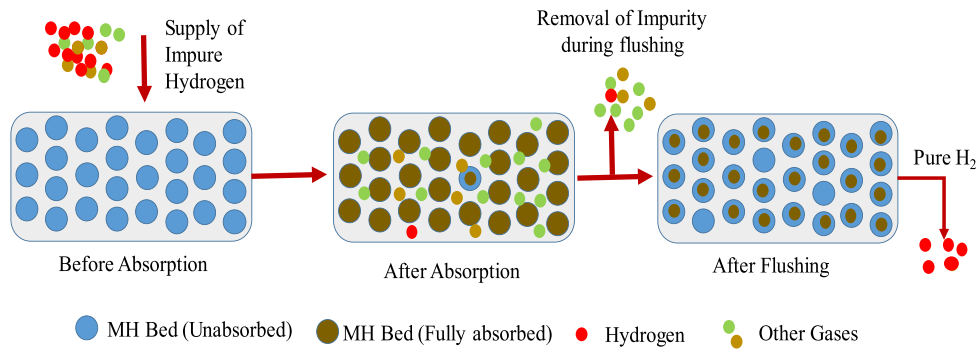


Fig. 1. Illustration on working of metal hydride based hydrogen purification system (MHPS).

Table 1

Experimental parameters for the coupled reactor system study.

Type of Study	Reactor involved	Operating Parameters	Gas Composition
Parametric study on rate of hydrogen transfer in coupled reactor system	R1 to R2 R1 to R3 R2 to R3	Desorbing reactor temperature (T _d): 70 °C, 80 °C & 90 °C Absorbing reactor temperature (T _a): 20 °C & 30 °C • R1: P _s = 20 bar & T _a = 20 °C • R1 to R2: T _d = 90 °C & T _a = 20 °C • R2 to R3: T _d = 90 °C & T _a = 20 °C • R3: T _d = 90 °C	Pure Hydrogen (99.995% purity)
Multistage hydrogen purification study	R1, R2 and R3		Hydrogen with 80% impurity (CO, CO ₂ , CH ₄ , N ₂ and Ar)

directly depends on purity level of hydrogen. But, hydrogen obtained through most of the production techniques are not pure [19]. Hence, hydrogen purification plays a significant role in the overall hydrogen value chain. Some well established hydrogen purification techniques, such as pressure swing adsorption, membrane diffusion, cryogenic separation, metal hydride (MH) systems, catalytic purification bases purification etc. [19–23], are available for H₂ separation from impure gas mixtures. Among these techniques, metal hydride based hydrogen separation is the simplest and economical technique to purify hydrogen using thermal heat input (Fig. 1), wherein, hydrogen separation is accomplished in three steps, i.e. absorption, flushing and desorption [24]. During absorption, impure gas mixture is supplied to the metal hydride (MH) reactor which is maintained at low temperature through regular circulation of heat transfer fluid (HTF). As metal hydride gas high affinity towards hydrogen, it absorbs only hydrogen and the other gaseous impurities remains in the void volume. Absorption is followed by flushing, which is performed at low temperature and the impure gases present in the void space is released to the atmosphere. In the last stage i.e. desorption, the temperature of MH bed is raised to elevated temperature, and pure hydrogen is desorbed as an end product.

Metal hydride reactor is an engineered vessel with sufficient mechanistic strength, that allows the intermetallic alloy to remain in tacked, and facilitate the hydrogen transfer to/from the MH reactor and simultaneously assists the smooth flow of HTF through the bed to

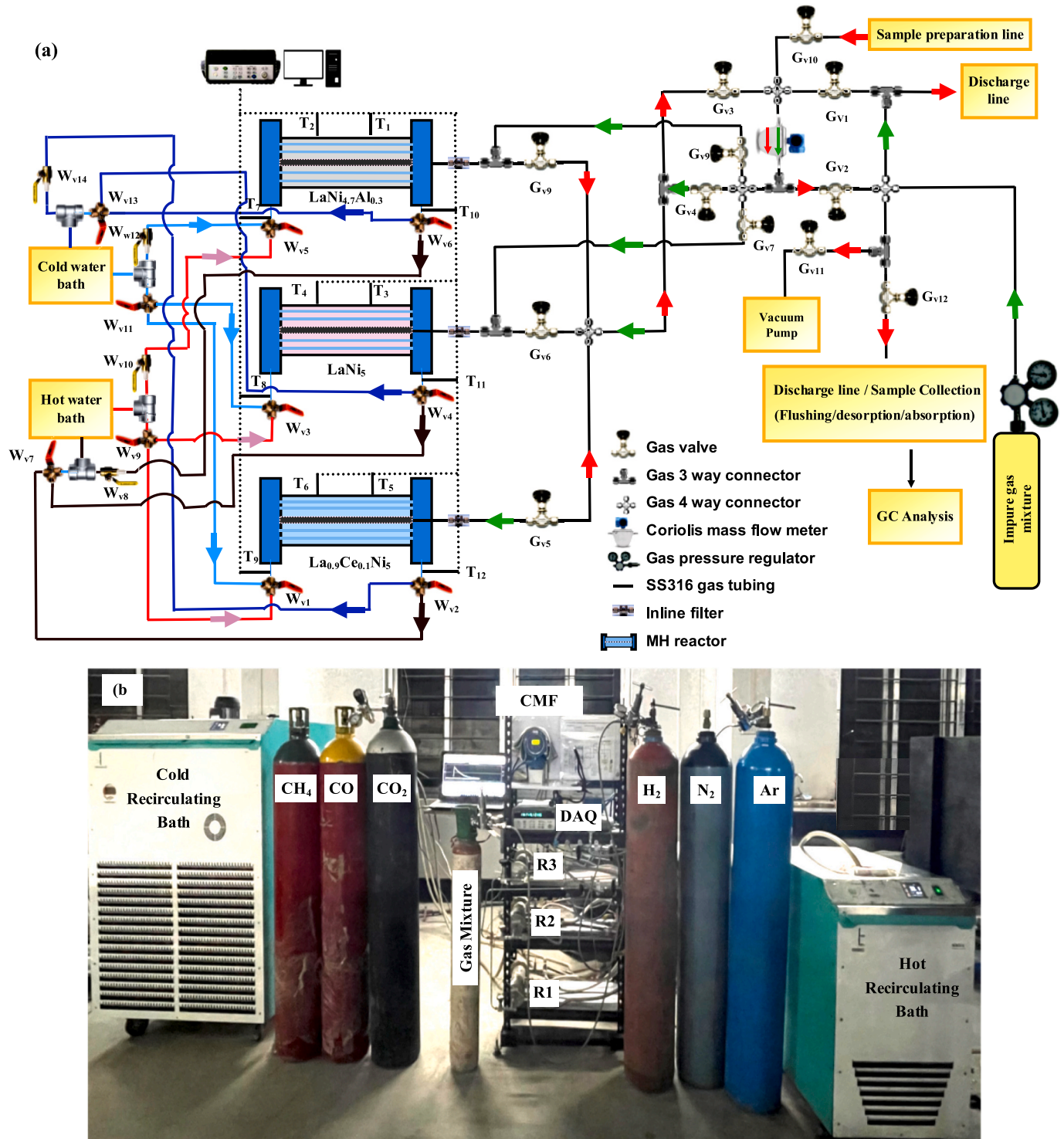


Fig. 2. (a) Detailed schematic of 3-Stage metal hydride based hydrogen purification system with HTF and Gas circuit and (b) pictorial view of the 3-stage metal hydride based hydrogen purification system

accomplish hydriding/dehydriding process. The MH reactor is designed on the principle of optimization of combined heat and mass transfer characteristics of the MH bed. As the thermal conductivity of MH alloys are very less [25,26], it becomes very essential to optimize bed thickness in such a manner, that it could facilitate effective transfer of excess heat from the bed, in order to boost its absorption/desorption kinetics [27,28].

In past, several studies have been performed on the reactor design, wherein parameters like reactor strength, hydrogen transfer, bed

packing, reactor configuration, fabrication feasibility etc., are considered while designing a MH reactor [28–30]. Alloy selection in MH-hydrogen system also plays a significant role in executing the targeted applications successfully and efficiently [31]. Pressure-Concentration-Isotherm (PCT) of MH alloys can be tailored, by altering its intermetallic composition according to its application [24,32]. Some of the alloys like MgH_2 , complex hydrides, etc., have comparatively high H_2 storage capacity (~ 7.5 wt%), but due to their high stability and high desorption temperatures, these are not suitable for MHHPS [33,34].

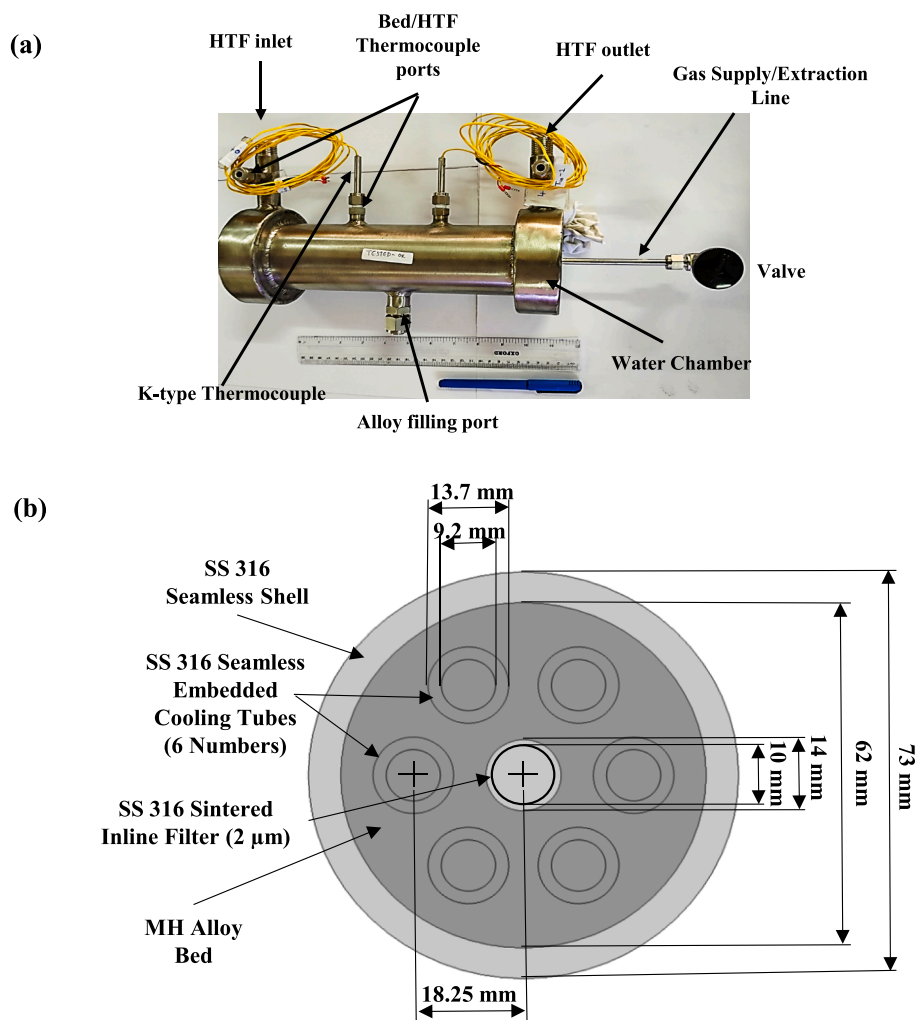


Fig. 3. (a) Pictorial view of the reactor and (b) reactor configuration with dimensions.

However, some metal hydrides fall under the group of AB_5 , A_2B and AB_2 are suitable for hydrogen purification application [19,35].

In 1996, Au et al. [36] reported the use of metal hydride ($MNi_{1.5-x}Al_x$) for storing, purifying, recovering, and transporting hydrogen from a purge gas line of an ammonia plant. The system reported hydrogen recovery of around 70% with up to 97% saving in power consumption, which resulted to 60% saving in hydrogen price. According to Dunikov et al. [37], N_2 hardly affects purification characteristics of AB_5 , but it decreases storage capacity by 3–4%. A test was performed with 27% N_2 in $H_2 + N_2$ gas mixture, which was fed to $LaFe_{0.1}Mn_{0.3}Ni_{4.8}$. Hydrogen loss of <5% was observed during the recovery phase. However, impurities such as CO, CO_2 and H_2S significantly affect rate of reaction by slowing down initial rate of reactor by 2 to 20 times [38,39]. According to Oztek M.T [40], impurities like He and Ar have no impact on the reversible purification characteristics of alloys like Mg_2Ni , $VTiNi$ and $LaNi_5$. However, the impurities like CO_2 , CO, O_2 , H_2S can cause alloy poisoning. Handa et al. [41] reported poisoning characteristics of CO_2 on various AB_5 alloys. The poisoning characteristics of CO_2 on various AB_5 alloys, varied in the sequence $CaNi_5 < LaNi_5 < MnNi_{4.025}Co_{0.4}Mn_{0.275}Al_{0.3} < LaCo_5$ respectively. Compared to other gaseous impurities, the poisoning effect of CO was comparatively more dominant. Wan et al. [42] reported poisoning characteristics of CO on $LaNi_{4.7}Al_{0.3}$, wherein H_2 with 300 ppm CO content was used for the study. Significant poisoning was observed, and the alloy got deactivated after 3 cycle, wherein absorption was carried out at 30 °C. But the storage capacity was not affected, when tested at 80 °C. However, the poisoned alloy was

regenerated/reactivated after one pure hydrogen absorption cycle at high pressure.

Use of intermetallic alloys for bio hydrogen purification and storage has been proven as an efficient technology in past two decades [43–45]. However, depending upon the content of the other gases in impure hydrogen sample, purification process may be accomplished in single or multiple purification stages [46,47]. Kazakov et al. [46] reported the use of $LaNi_{4.4}Fe_{0.3}Al_{0.3}$ and $LaNi_{4.6}Mn_{0.2}Al_{0.2}$ as hydrogen purification alloy, to separate hydrogen from gas sample of biological hydrogen production stream. They reported that the alloys were suitable for hydrogen separation from a dilute gas stream from biohydrogen production line. Similarly, Dunikov et al. [47] reported the hydrogen separation from a biohydrogen production sample with concentration in the range of 40–60 vol% and feed pressure of 5.6 bar. The study reported 2-step purification using $La_{0.9}Ce_{0.1}Ni_5$ and $LaNi_{4.8}Mn_{0.3}Fe_{0.1}$ alloy (1 kg each) in each purification step. The purified hydrogen was sent to PEM fuel cell, and it was observed that, the degree of purification was sufficient to run a 200 W PEM fuel cell to its maximum capacity. Dunikov and Blinov [48–50] studied hydrogen separation from lean mixture of CH_4-H_2 (90% CH_4 and 10% H_2), wherein the supply pressure of mixture was 9.5 bar and the alloy used was $LaNi_{4.8}Mn_{0.3}Fe_{0.1}$. Though the study does not report the purification level of the hydrogen obtained, but reported the hydrogen recovery in lean mixture, which drop down to ~74% in the present case. AB_2 alloys such as $Zr_{0.9}Ti_{0.1}V_2$ [51], $Ti_{0.515}Zr_{0.485}Cr_{0.8}$ [52] and $Ti_{0.515}Zr_{0.485}Mn_{1.2}Mo_{0.1}$ [52] can also be used for hydrogen purification as an alternative to AB_5 alloys. According to Hanada et al.

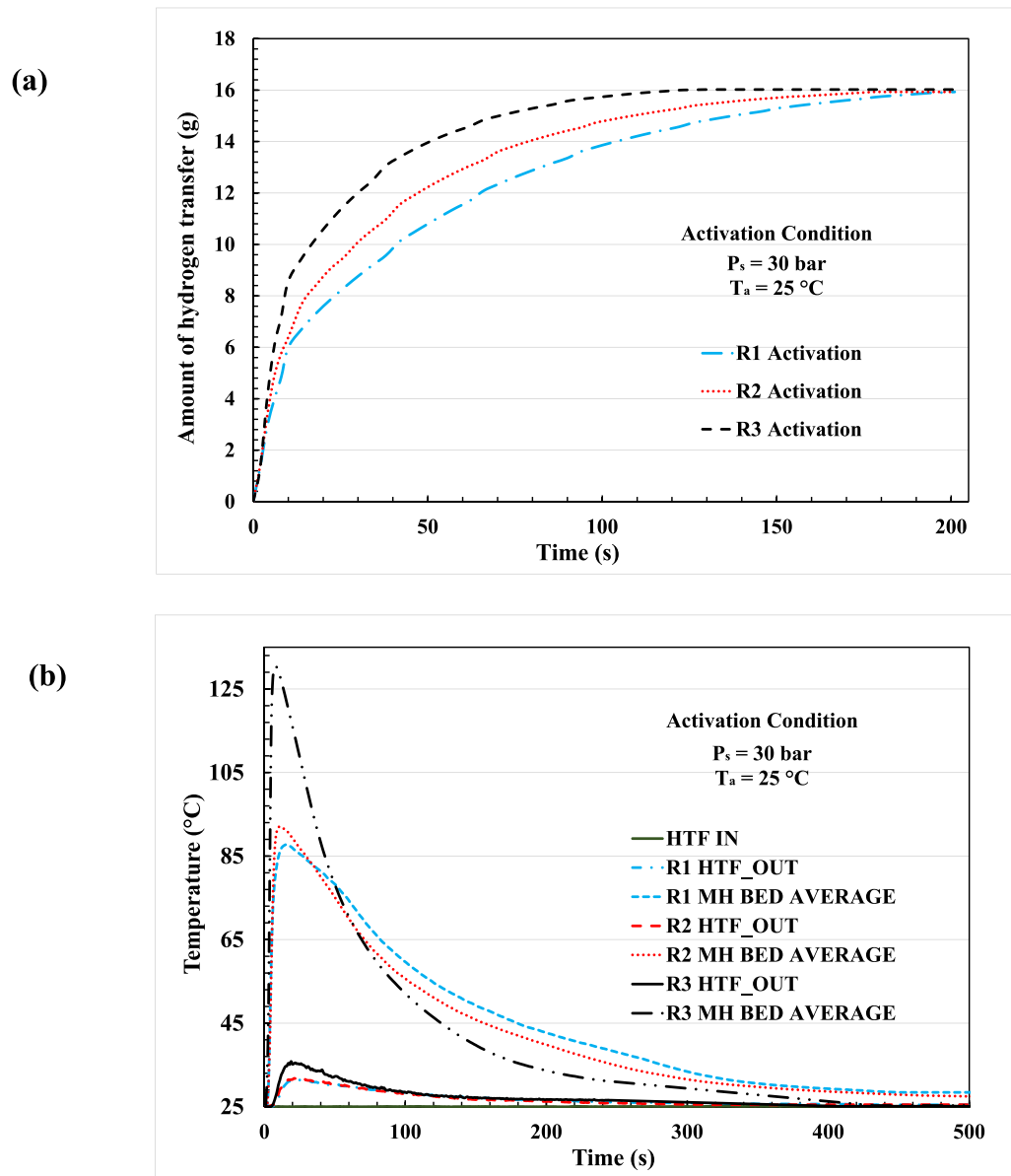


Fig. 4. Plot for (a) amount of H_2 absorbed and (b) HTF and MH bed temperature variation during activation cycle of reactors R1 ($\text{La}_{0.9}\text{Ce}_{0.1}\text{Ni}_5$), R2 (LaNi_5) and R3 ($\text{LaNi}_{4.7}\text{Al}_{0.3}$).

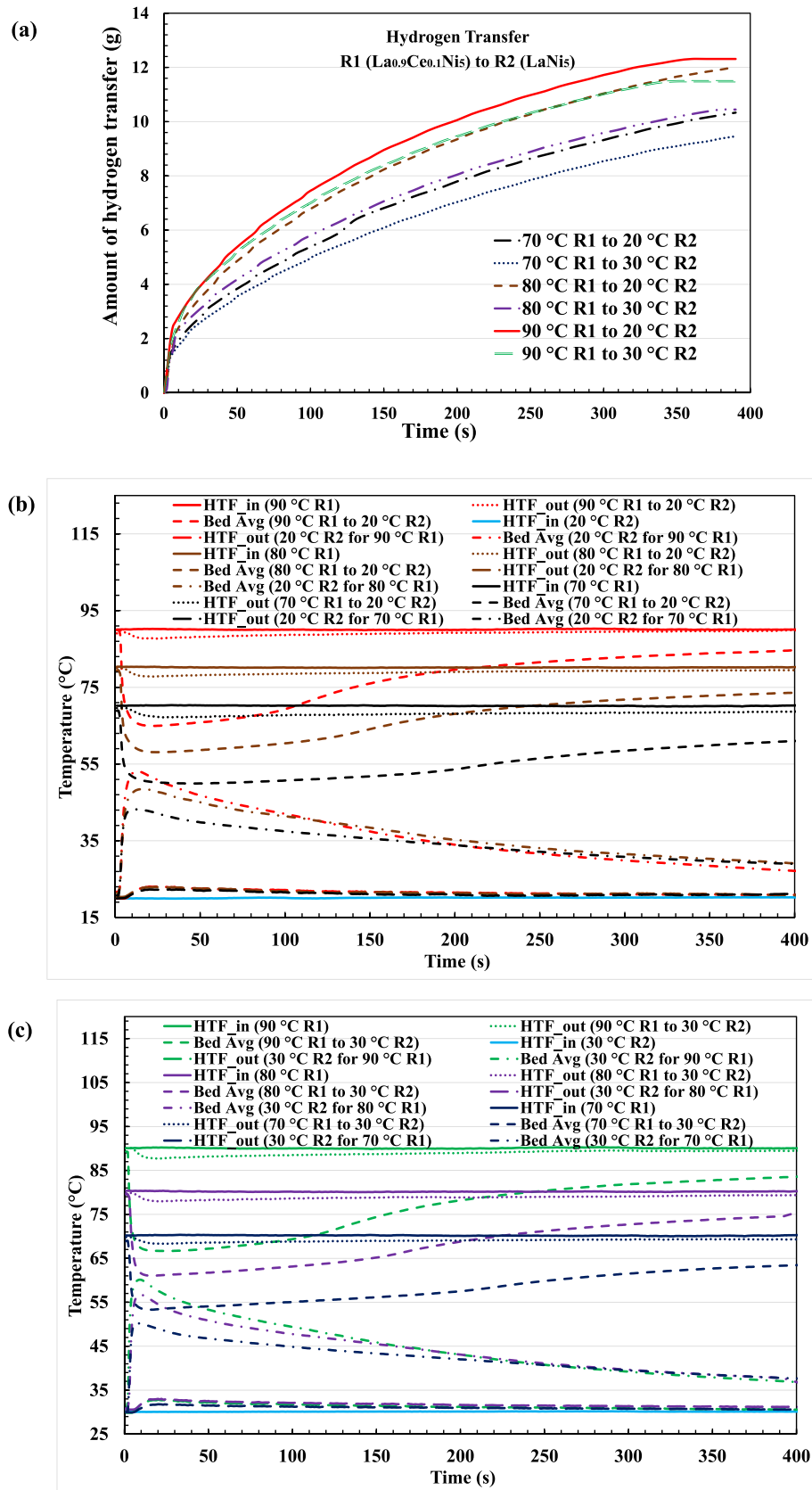


Fig. 5. (a) Rate of hydrogen transfer from R1 to R2 at different absorption/desorption temperatures, (b) temperature variation curve of desorbing reactor (R1) to absorbing reactor (R2) at 20 °C and (c) temperature variation curve of desorbing reactor (R1) to absorbing reactor (R2) at 30 °C.

Table 2
Details of hydrogen transfer in the coupled reactor (R1 to R2).

T _d (R1)	T _a (R2)	H ₂ Transfer from R1 to R2	wt%	Time
70 °C	20 °C	10.34 g	0.86	390 s
70 °C	30 °C	9.46 g	0.79	385 s
80 °C	20 °C	12.02 g	1.01	390 s
80 °C	30 °C	10.45 g	0.87	380 s
90 °C	20 °C	12.31 g	1.03	350 s
90 °C	30 °C	11.49 g	0.96	350 s

[52], the order of poisonings tolerance of some AB₂ alloys are Ti_{0.515}Zr_{0.485}Mn_{1.2}Fe_{0.1} < AB₂-Ni < AB₂ < AB₂-Co < AB₂-Fe (AB₂ = Ti_{0.515}Zr_{0.485}Cr_{0.8}) respectively.

Though hydrogen purification using intermetallic alloys is a known technology, however obtaining ultrapure hydrogen (99.9995%) from a sample with higher impurity range (50–80%) using low grade heat input (80–90 °C) is still a challenge. There is a significant research gap in exploring the right combination of metal hydride alloys for hydrogen purification from higher level of impurity. Considering the scope of research, a sustainable multistage thermally driven metal hydride based hydrogen purification system was studied and reported in the present work. The choice of multistage purification technique for the present study facilitates the system ability to handle higher range of impurity content in the supply sample. The MH alloys are selected based on their thermodynamic properties, such that the system can work in low temperature and pressure range. The consideration of low working temperature allows futuristic approach for its coupling with waste heat recovery or solar thermal systems having low grade heat input (up to 90 °C). Moreover, the multistage purification technique will facilitate the recovery of ultrapure hydrogen from the gas samples with even higher level of impurity [53].

2. Methodology

Three identical MH reactors with 6 embedded cooling tube (ECT) were fabricated using SS316 material and each were filled with 1.2 kg La_{0.9}Ce_{0.1}Ni₅, LaNi₅, and LaNi_{4.7}Al_{0.3} in the shell space of each reactor, and were placed in first, second and third purification stages, respectively. Water as HTF was used, that flows through ECT of the reactor, facilitating effective heat transfer within the reactor. In order to obtain suitable parameters for the MHPS operation, parametric studies on coupled reactor system was performed by varying absorption and desorption parameters, which is depicted in Table 1, wherein pure hydrogen was used for study. A complete one cycle of H₂ separation process consists of three steps i.e. absorption of gas mixture, flushing of impure void gases and desorption of H₂ absorbed. The suitable operating parameters for the absorption, flushing and desorption were obtained through parametric studies.

During absorption, reactor was maintained at low temperature (20 °C) and the impure gas mixture was sent through the gas circuit to the reactor at 20 bar. During absorption, H₂ gets absorbed in the intermetallic alloys forming hydride (solid solution), whereas the impurity remains in the void space. Due to exothermic absorption, HTF recirculation at room temperature is facilitated through ECT of the reactor, to remove the exothermic heat of reaction, from the MH bed to ensure faster absorption kinetics. Further, the reactor is flushed at low temperature (~15 °C) to ensure removal of impurity without losing hydrogen. Further, reactor is prepared for desorption, wherein it is raised to 90 °C through continuous HTF recirculation through high temperature recirculating bath. Once the reactor attains the desired temperature, the desorption circuit is opened to collect the purified hydrogen. A Coriolis mass flow meter installed in the gas circuit records the amount of gas transfer to/from the reactor during absorption, flushing and desorption. The single and multistage operation of the purification system is determined by level of impurity present in the gas

sample. For higher impurity content (>20%), the system is operated in multi stage. The schematic and pictorial view of the test setup is depicted in Fig. 2a and Fig. 2b.

2.1. Alloy and reactor details

Initially, experiments on the coupled reactor system was performed, wherein three identical reactors with 6 embedded cooling tubes (ECT) were filled with different AB₅ alloys and arranged as multiple hydrogen purification batteries/bed in series. The pictorial view of the 6 ECT reactor with its detailed specification and cooling tube arrangement is depicted in Fig. 3. The basic configuration of the reactor is similar to that of shell and tube type heat exchanger, wherein the shell side MH alloy is placed, and the ECT's facilitates the effective heat transfer to/from the MH bed. Stainless steel sintered filter of 2 µm pore size is welded at the centre of the reactor to facilitate uniform hydrogen absorption/desorption process along the axial direction, keeping alloy in tacked within the reactor shell. These alloys were arranged in the descending order of their equilibrium bed pressure at room temperature (PCT characteristics), such that 'stage 1 reactor (R1)' was filled with La_{0.9}Ce_{0.1}Ni₅, 'stage 2 reactor (R2)' with LaNi₅ and 'stage 3 reactor (R3)' with LaNi_{4.7}Al_{0.3} and these reactors were filled with equal alloy mass of 1.2 kg each. The reason for arranging the metal hydride alloys in its descending order of PCT characteristics is because of purification application. Considering the operational temperature and pressure ranges, the descending order (PCT) of material arrangement, facilitates maximum hydrogen transfer during coupling, due to higher pressure gradient for a given temperature range. Parallely, the last stage alloy, being more stable as compared with adjacent stages (low equilibrium pressure), ensures ultrapure hydrogen delivery because of efficient flushing.

Initially the reactors and the gas circuit were tested at 100 bar pressure, to ensure the leak proof in gas circuit. The leak test was performed using argon gas. Before conducting coupled system studies, activation of intermetallic alloys was performed by conducting successive hydrogenation/dehydrogenation cycles for all the reactor beds. The activation was carried out using pure hydrogen, which was supplied at 30 bar, wherein the absorption temperature (HTF inlet) of 25 °C was maintained. The absorption half cycle was followed by the desorption half cycle which was performed at 90 °C, and the reactor was further evacuated down to 10⁻² mbar, before the next cycle. The reactors were reached their maximum absorption capacities and got activated in the third activation cycle, wherein R1, R2 and R3 absorbed 15.91 g, 15.93 g and 16.02 g respectively in 201 s, 178 s and 130 s (Fig. 4a).

The given absorption amount contributes to 1.32 wt%, 1.33 wt% and 1.34 wt% of the hydrogen storage capacity for respective reactors. The corresponding peak bed temperature achieved for each of R1, R2 and R3 were 87.8 °C, 92 °C and 130.5 °C respectively in 16th s, 11th s, and 8th s. As depicted in Fig. 4b, their respective HTF outlet temperature was 31.4 °C in 22 s, 31.7 °C in 21 s and 35.5 °C in 22 s, for 25 °C absorption temperature and 30 bar H₂ supply pressure. During studies, water as HTF at fixed flow rate of 4 lpm was used. The uncertainty in experimental measurements was calculated using method adopted by Kline and McClintock [54]. The uncertainty in measurement was ±2.4%, ±0.5 °C and ± 2% respectively for hydrogen storage, temperature and pressure measurement. The impure hydrogen mixture was prepared using Coriolis mass flow meter, which had flow accuracy of ±0.25%.

2.2. Absorption of impure mixture

Initially, impure gas mixture was sent to stage 1 reactor (R1). Absorption was performed at 20 °C and 20 bar gas supply pressure. The temperature was maintained by constant supply of water as HTF through ECT of reactor. During this recirculation, HTF (water) valves W_{v1}, W_{v2}, W_{v11} and W_{v14} were kept in open position and cold temperature bath in running mode (Fig. 2a). The impure mixture was supplied

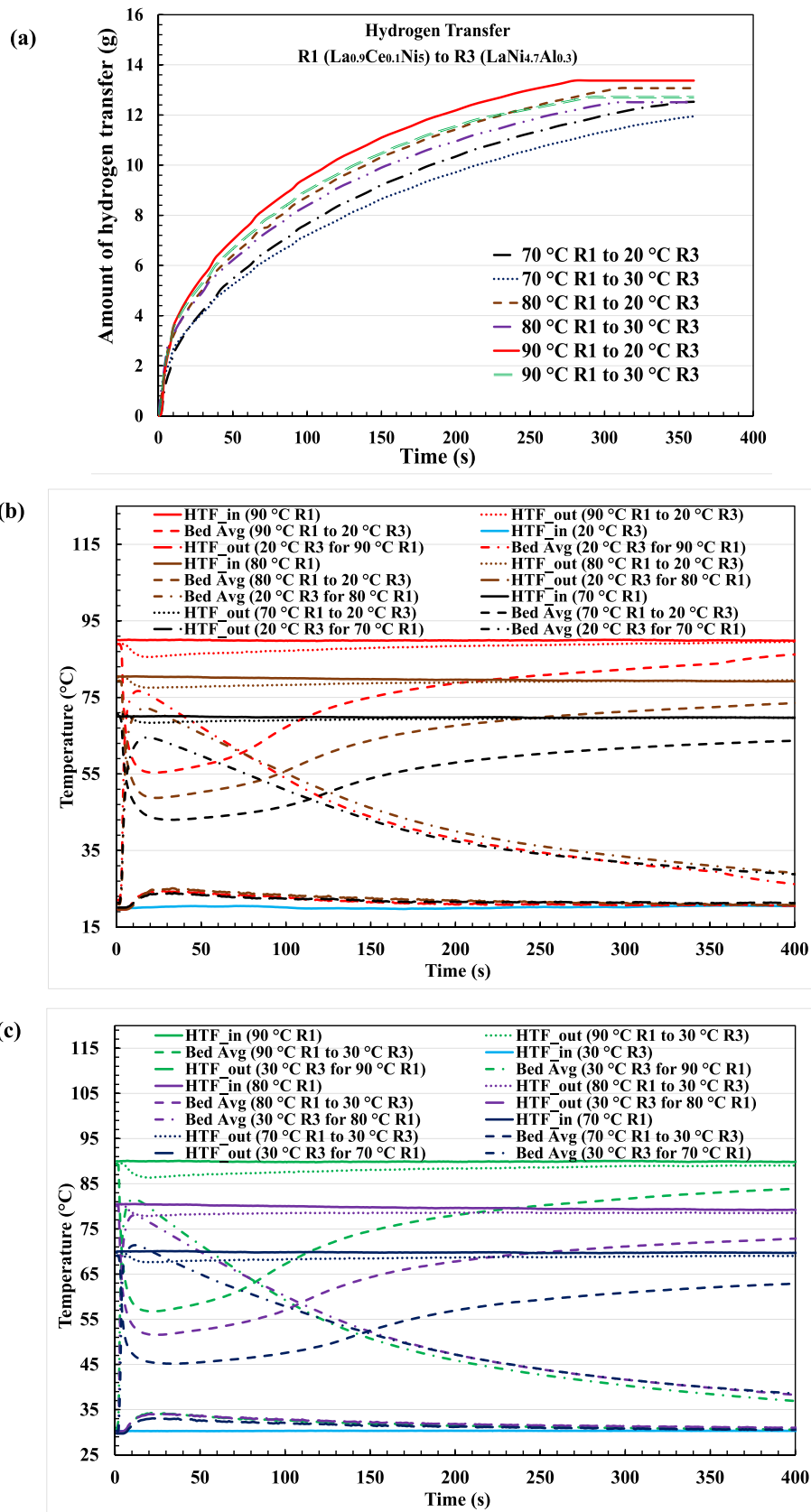


Fig. 6. (a) Rate of hydrogen transfer from R1 to R3 at different absorption/desorption temperatures, (b) temperature variation curve during hydrogen transfer from R1 to R3 at 20 °C and (c) temperature variation curve during hydrogen transfer from R1 to R3 at 30 °C.

Table 3

Details of hydrogen transfer in the coupled reactor (R1 to R3).

T _d (R1)	T _a (R3)	H ₂ Transfer from R1 to R3	wt%	Time
70 °C	20 °C	12.53 g	1.04	360 s
70 °C	30 °C	11.95 g	0.99	360 s
80 °C	20 °C	13.07 g	1.08	310 s
80 °C	30 °C	12.51 g	1.04	305 s
90 °C	20 °C	13.38 g	1.12	280 s
90 °C	30 °C	12.71 g	1.06	290 s

to stage 1 reactor through gas circuit, wherein valve G_{v1}, G_{v4} and G_{v5} were kept open and supply cylinder was set to 20 bar using pressure regulator. Further, gas mixture uniformly spreads within the reactor through a sintered porous tube filter of 2 µm pore size. After coming in contact with the gas mixture, R1 containing La_{0.9}Ce_{0.1}Ni₅, starts absorbing hydrogen to form La_{0.9}Ce_{0.1}Ni₅-H₆. At the same time, the other gaseous impurities remain unabsorbed in the void space of the reactor. Due to exothermic nature of absorption reaction, continuous supply of low temperature HTF is maintain to remove the excess heat of reaction to ensures faster absorption rate.

2.3. Flushing of void gases

In order to remove unabsorbed impure gases from the void space, reactors were flushed at low temperature (15 °C). During flushing, the recirculating chillers were brought down to 15 °C and HTF was recirculated through the MH reactor. During recirculation, HTF valve positioning was same as absorption process, however the gas circuit was set similar to desorption circuit, wherein, valves G_{v5}, G_{v3} and G_{v2} were kept open and impure gases were directly flushed to atmosphere, or removed through vacuum pump. The flushing process is completed in a few seconds (5–10 s), as the void volume was less. For bigger reactor systems, it may take longer time to flush the reactor. After flushing, the gas valves were closed, and the reactor was prepared for desorption process.

2.4. Desorption of the absorbed hydrogen

After performing flushing, reactor was prepared for desorption, wherein HTF at 90 °C was recirculated through R1 using hot recirculating bath. For facilitating HTF recirculation, valves W_{v1}, W_{v2}, W_{v7} and W_{v9} were kept open, keeping recirculation pump in on position. Simultaneously, the absorbing reactor R2 was maintained at low temperature (20 °C), through continuous recirculation of HTF using recirculating chillers, keeping valves W_{v3}, W_{v4}, W_{v11} and W_{v13} in open position. Once thermal equilibrium between MH bed and HTF for the respective absorbing (R2) and desorbing (R1) reactor was maintained, hydrogen from R1 was sent to R2. During this process, gas valves G_{v5}, G_{v3} and G_{v7} was kept opened to facilitate gas transfer from R1 to R2. Once, the transfer was completed, R2 was further prepared for flushing (cooled down to 15 °C), and once the flushing temperature was attained, the gas valves G_{v6}, G_{v3} and G_{v2} were opened to remove void gases.

Further, to obtaining ultra-pure grade of hydrogen (~99.9995% purity), hydrogen transfer from R2 to R3 was performed, wherein R2 was heated to 90 °C and R3 was cooled to 20 °C, by keeping respective water valves in open position. Once the thermal equilibrium between the MH bed and HTF was attained, gas valves G_{v6}, G_{v3} and G_{v9} were opened to facilitate H₂ transfer from R2 to R3. At end R3 was flushed at 20 °C, similar to flushing in R1 and R2, keeping gas valves, G_{v8}, G_{v3} and G_{v2} in open position and facilitating recirculation of cold HTF through W_{v5}, W_{v6}, W_{v12} and W_{v14}. Finally, the ultra-pure H₂ was obtained after desorption of R3. During desorption, R3 was raised to 90 °C using HTF recirculation through valves W_{v5}, W_{v6}, W_{v10} and W_{v9}. Once the bed attains thermal equilibrium with HTF, the gas valve G_{v8}, G_{v3} and G_{v2} was opened to collect ultra-pure H₂. During hydrogen purification process, the temperature variation in the MH bed and HTF was recorded

using K-type thermocouples, which was connected to a data acquisition system (DAQ). The gas circuit was completely isolated from atmosphere and the system operated in a close loop. The hydrogen transfer to/from the reactor were recorded using the Coriolis mass flow meter (CFM).

After certain number of purification cycles, the bed poisoning was witnessed. Therefore, the MH bed was regenerated time to time (depending upon impurity level). The MH bed was regenerated by repeating activation process, wherein the MH reactor was evacuated down to 10⁻² mbar at 95 °C, using a vacuum pump and recirculation of high temperature HTF. Once the bed achieves set vacuum pressure, all the valves were closed and the reactor was cooled down to perform absorption using pure hydrogen. During the regeneration absorption process, pure hydrogen was supplied at higher pressure (20–30 bar). Once the absorption was completed, MH bed was again raised to higher temperature (95 °C) and the evacuation process was repeated followed by absorption. This absorption-desorption-evacuation was repeated till the bed gets regenerated. It takes 2–3 cycle for complete regeneration of the MH bed.

2.5. Gas analysis

The gas samples during absorption, desorption and flushing were collected and analysed using TCD analysis in gas chromatography. CP Molesieve 5A column was used for the analysis, wherein oven temperature was raised from 40 to 150 °C to facilitate gas separation within the column. Initially the oven was maintained at 40 °C for 10 min and further, the temperature of oven was raised to 150 °C in a time step of 20 °C / min. The retention time of the peak was detected using pure gas samples of different gases. It was observed that, for column flow rate of 6 lpm, the H₂, Ar, N₂, CH₄ and CO peaks were detected at retention time of 3–4th min, 5th min, 8–9th min, 11th min and 15–16th min respectively, considering above oven program. Hydrogen showed a negative peak because of lower thermal conductivity of He (carrier gas) as compared to hydrogen. Once peak oven temperature was achieved, it was further continued for 10 min to facilitate gas detection. The detector temperature was fixed as 250 °C during the gas analysis. At end of each sample analysis, the column was further raised to 300 °C, to ensure no gases left out before the next sample get injected.

3. Results and discussions

This section presents the experimental outcomes of hydrogen transfer and temperature variation during coupled reactor study. Capability of metal hydride materials for hydrogen separation from lean hydrogen mixture are also discussed along with TCD results of gas samples at various stages of hydrogen purification.

3.1. Heat and mass transfer in coupled reactor system

In order to depict the suitable operating parameters for the efficient working of coupled reactor system, the parametric study was performed by varying absorption/desorption operating conditions as depicted in Table 1. The parametric studies were performed using commercial grade type-1 hydrogen cylinder with 99.9995% purity.

3.1.1. Hydrogen transfer from R1 to R2

In this study, the desorbing reactor (R1) was filled with a fixed amount of 14.5 g hydrogen while the absorbing reactor (R2) was kept empty. From the result obtained (Fig. 5a), it was observed that, the rate of hydrogen transfer was largely affected by the operating temperatures of absorbing and desorbing reactors. For 70 °C desorption temperature (R1) and 30 °C absorption temperature (R2), the amount of hydrogen transfer from R1 to R2 was 9.46 g, which was just 65.3% of the total hydrogen available for transfer in R1.

With increase in the temperature of desorbing reactor and decrease in the temperature of absorbing reactor, hydrogen transfer from R1 to

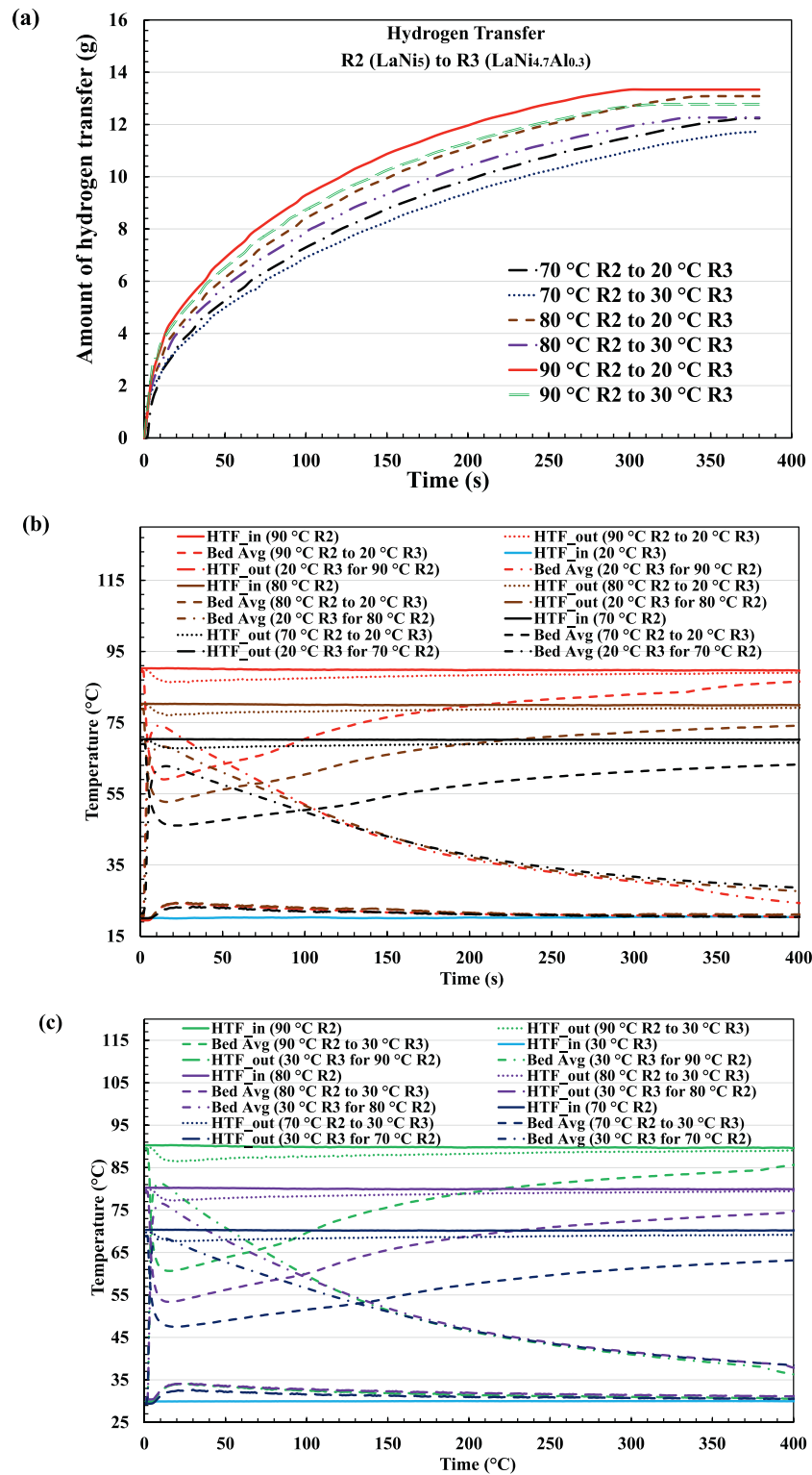


Fig. 7. (a) Hydrogen transfer from R2 to R3 at different absorption/desorption temperatures and (b) temperature variation curve during hydrogen transfer from R2 to R3 at 20 °C and (c) temperature variation curve during hydrogen transfer from R2 to R3 at 30 °C.

R2 was increased significantly. For 90 °C desorption temperature (R1) and 20 °C absorption temperature (R2), the amount of hydrogen transfer from R1 to R2 was 12.31 g, which constitutes to nearly 85% of the total hydrogen available for transfer. The exothermic and endothermic nature of the absorption/desorption reaction in the coupled reactor system can be observed by the temperature variation within the MH bed and HTF during coupled reactor study (Fig. 5b & Fig. 5c). With increase in

temperature difference between the absorbing and desorbing reactors, the pressure gradient between the reactors also increases, which leads to faster rate of hydrogen transfer between the reactors. Moreover, for lower absorption temperature, the equilibrium bed pressure of the alloy is lower, this increases the rate of absorption for the absorbing reactor. The details of the hydrogen transfer between the coupled reactor R1 & R2 is depicted in Table 2.

Table 4
Details of hydrogen transfer in the coupled reactor (R2 to R3).

(T _d) R2	(T _a) R3	H ₂ Transfer from R2 to R3	wt%	Time
70 °C	20 °C	12.24 g	1.02	370 s
70 °C	30 °C	11.72 g	0.98	380 s
80 °C	20 °C	13.08 g	1.09	350 s
80 °C	30 °C	12.26 g	1.02	340 s
90 °C	20 °C	13.33 g	1.11	300 s
90 °C	30 °C	12.77 g	1.06	320 s

3.1.2. Hydrogen transfer from R1 to R3

Similarly, the experiments were performed to study the reaction kinetics of hydrogen transfer from R1 to R3 as coupled system, wherein temperatures of the reaction bed were varied as per selected parameters (Table 1). The rate of hydrogen transfer was directly proportional to the difference between the absorbing and desorbing bed temperature. As depicted in Fig. 6a, the rate and amount of hydrogen absorbed was less for 70 °C desorption temperature (R1) and 30 °C absorption temperature (R3), and it was maximum for 90 °C desorption temperature (R1) and 20 °C absorption temperature (R3). The amount of hydrogen transfer varied from 11.948 g in 360 s to 13.37 g in 280 s, which was in the range of 82.4–92.2% of the total hydrogen available for transfer. As shown in Table 3, for the various operating temperatures, the reaction kinetics got varied, which was mainly because of the difference in the equilibrium bed pressure of the absorbing and desorption bed at different temperature. The endothermic/exothermic reactions can be clearly observed in Fig. 6b and Fig. 6c, wherein the temperature variation plot for the coupled reactor system is presented. Due to significant pressure difference for lower (20 °C) absorbing and higher (90 °C) desorbing MH bed (Fig. 6b), the bed temperature variation has showed highest rise and drop for the respective absorbing and desorbing beds.

3.1.3. Hydrogen transfer from R2 to R3

Similarly, hydrogen transfer between R2 and R3 was also studied. For 70 °C desorption temperature (R2) and 30 °C absorption temperature (R3), the amount of hydrogen transfer was 11.72 g, which was 80.8% of the total hydrogen available for transfer. Hydrogen transfer from R2 to R3 increased significantly with increase in the temperature of desorbing reactor and decrease in the temperature of absorbing reactor (Fig. 7a). For 90 °C desorption temperature (R2) and 20 °C absorption temperature (R3), the amount of hydrogen transfer was 13.33 g, which constitutes of nearly 92% of the total hydrogen available for transfer. With increase in temperature difference between the absorbing and desorbing reactors, the pressure gradient between the reactors also increases. This leads to faster rate of hydrogen transfer between the reactors. Moreover, for lower absorption temperature, the equilibrium bed pressure of the alloy is lower, this increases the rate of absorption for the absorbing reactor. The details of the hydrogen transfer between the coupled reactor R1 & R3 is depicted in Table 4. The temperature variation of the MHHPS and HTF during coupled reactor study is depicted in Fig. 7b and Fig. 7c. The endothermic nature of the desorbing bed and the exothermic reaction in the absorbing bed can be clearly observed. For all the three cases, it was observed that for 90 °C desorption temperature and 20 °C absorption temperature, the amount of hydrogen absorbed was maximum. This is because of higher pressure difference between the equilibrium pressure of desorbing reactor at 90 °C and absorbing reactor at 20 °C. Therefore, it is suggested to maintain maximum temperature difference between the absorbing and the desorbing reactor.

3.2. Hydrogen separation from impure gas mixture

To test the developed multistage hydrogen purification system with higher level of impurities, experiments were conducted with the multiple gas sample containing 80% impurities in the form of CH₄, N₂, Ar, CO₂, CO and 20% H₂. The reason of selecting high range of impurity was to

test MHHPS for harsh operational conditions. The purification was carried in 3 stages and for each stage, TCD analysis of gas sample was performed using Agilent make CP Molsieve 5A column, wherein Helium was used as carrier gas.

3.2.1. Stage 1 Purification

Initially, the TCD analysis on the absorption gas sample was conducted and result obtained was calibrated with calibration sample. As shown in Fig. 8a, the gas composition constitutes of 20% H₂, 18% Ar, 20% N₂, 20% CH₄, 20% CO₂ and 2% CO by weight. Lesser CO content in the impure gas mixture was considered because of its significant effect on poisoning characteristics of metal hydride [38]. It was also reported that, significant CO content in the impure gas mixture degrades the absorption kinetics of the system in few purification cycles [42]. After TCD analysis, the impure gas sample was sent for absorption to stage 1 reactor i.e. R1 (La_{0.9}Ce_{0.1}Ni₅), maintaining absorption condition of 20 bar and 20 °C. As depicted in Fig. 8b, the amount of hydrogen absorbed in stage 1 reactor was 10.59 g in 262 s, attaining the peak bed temperature of 60 °C in 23 s and peak HTF temperature of nearly 22.5 °C. Despite of high supply pressure (20 bar) the rate of absorption was on the slower side, because of the high impurity content in the absorption gas sample. After completion of absorption half cycle, the reactor R1 was, further cooled down to 15 °C, and prepared for flushing. During flushing, the reactor was maintained at 15 °C, and the discharge valve was opened to flush the reactor to ambient conditions. The amount of gas came out during flushing was recorded as 1.03 g, which leaves 9.56 g hydrogen available for transfer within reactor R1. The flushed sample was further analysed using online TCD.

As depicted in Fig. 8c, the gases detected in the flushed sample constitute of 4.65% H₂, 30.19% Ar, 31.88% N₂ and 33.28% CH₄. However, due to significant poisoning characteristics of CO, it goes through methanation to form methane and moisture, once it reacts with hydrogen in presence of Ni of La_{0.9}Ce_{0.1}Ni₅, and hence, it was not detected in the flushed sample [55]. However, with lesser CO content and high desorption temperature (above 80–85 °C), the reactor regenerates to absorb hydrogen in subsequent absorption cycle [42]. Apart from these gases, CO₂, must have been present in the desorption sample, but it was not detected during TCD analysis because of molecular sieve column used for the analysis. It can be observed that, during flushing, impurities present in the void space of the reactor leaves the reactor with very little amount of hydrogen i.e. 4.65%, which constitutes to 0.04 g of total gas flushed. Hydrogen coming out during flushing is mostly the unabsorbed hydrogen, which is present in the void space of the reactor along with impurities. After successful flushing of the impurities from R1, the reactor was prepared for desorption, wherein the reactor was heated to 90 °C.

Simultaneously, the stage 2 reactor (R2 filled with LaNi₅) was prepared for absorption. Before hydrogen transfer in the coupled reactor system, R1 was maintained at 90 °C and R2 was maintained at 20 °C. Once the respective bed and HTF temperatures stabilizes, the reactor valves were opened for facilitating hydrogen transfer from R1 to R2. The amount of hydrogen transferred from R1 to R2 was 8.91 g in 330 s, which constitute of 93.20% of total gas available for transfer. During this process, the peak bed temperature of 49.5 °C in 14 s and peak HTF temperature of nearly 22.2 °C was attained in the absorbing reactor R2 (Fig. 9a). However, because of endothermic desorption the bed and HTF temperature of R1 dropped to 70.8 °C and 87.6 °C respectively. Further, during gas transfer from R1 to R2, sample after stage 1 purification was collected and analysed. As depicted in Fig. 9b, it was observed that, after stage 1 purification, 91.54% pure hydrogen was obtained with impurity of 8.46%. The impurities comprise of 2.54% Ar, 2.14% N₂, 3.79% CH₄ and trace of 0.01% CO. In the TCD results of the flushed gas sample (Fig. 8c), CO was not present majorly because of methanation and very low flushing temperature. Due to very low flushing temperature, even the trace of CO might not have come out due to significantly low-pressure gradient. But, a trace of CO was observed in desorption

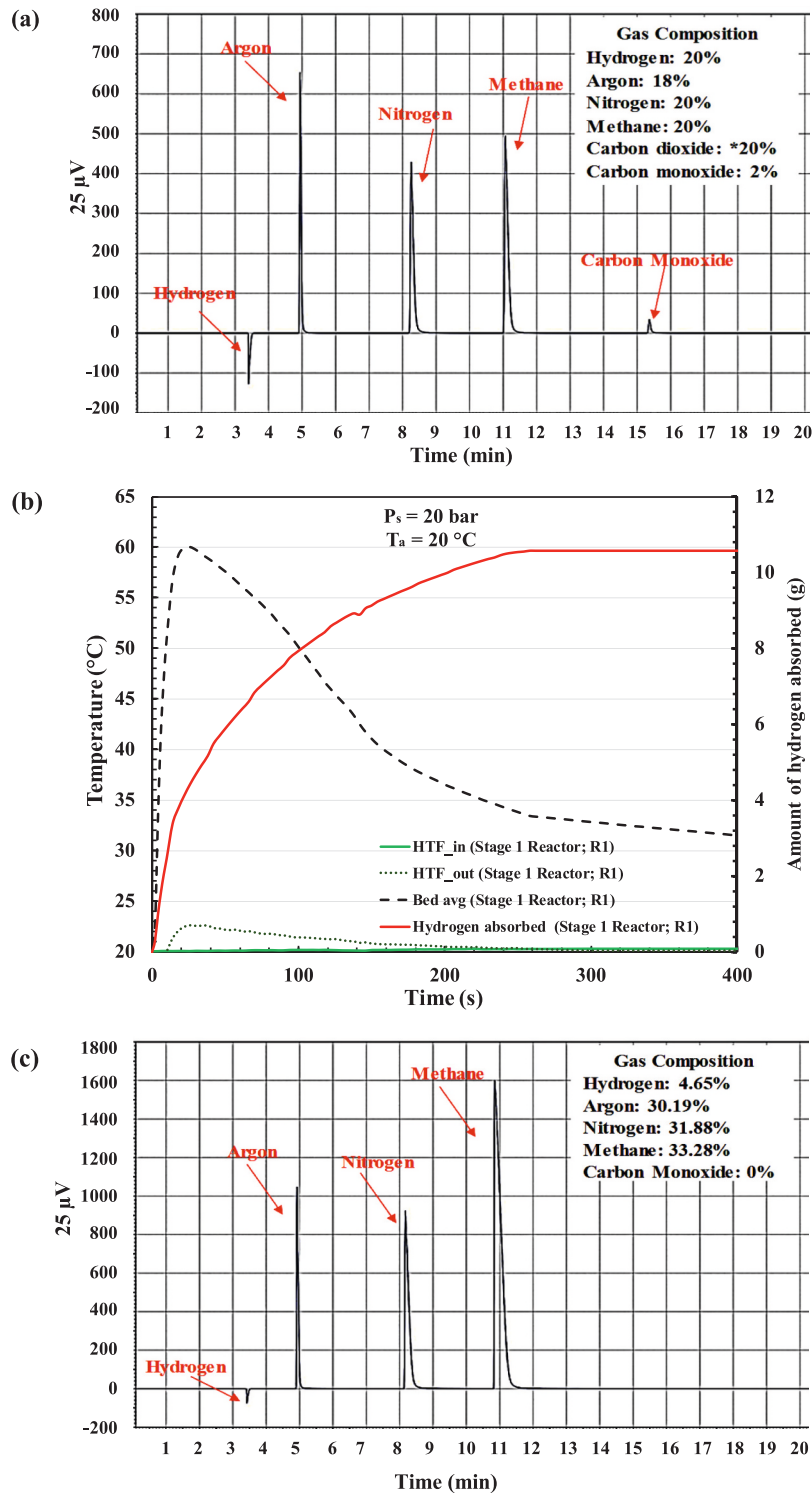


Fig. 8. (a) TCD results of the absorption gas mixture supplied to stage 1 reactor (R1) and (b) rate of gas transfer and temperature variation during absorption of impure gas sample to R1, and (c) TCD results of the flushed sample from stage 1 reactor.

*The TCD analysis was carried out using Agilent CP Molsieve 5A column. The column was unable to detect CO_2 as it gets absorbed at the wall of molecular sieve column, and does not elutes to detector.

sample after stage 1 purification. This was mainly because of high desorption temperature considered for efficient hydrogen transfer from desorbing reactor (at 90 °C) to absorbing reactor (at 20 °C). High desorption temperature, leads to discharge even smallest of gas packet from the reactor, due to significant pressure gradient between the desorbing and absorbing reactors. Moreover, to overcome the poisoning

effect of CO, high desorption temperature facilitates alloy regeneration in each desorption cycle [42]. The purity level of hydrogen obtained after stage 1 purification was low due to presence of lean hydrogen mixture as inlet sample, and methanation due to CO presence in the impure gas mixture.

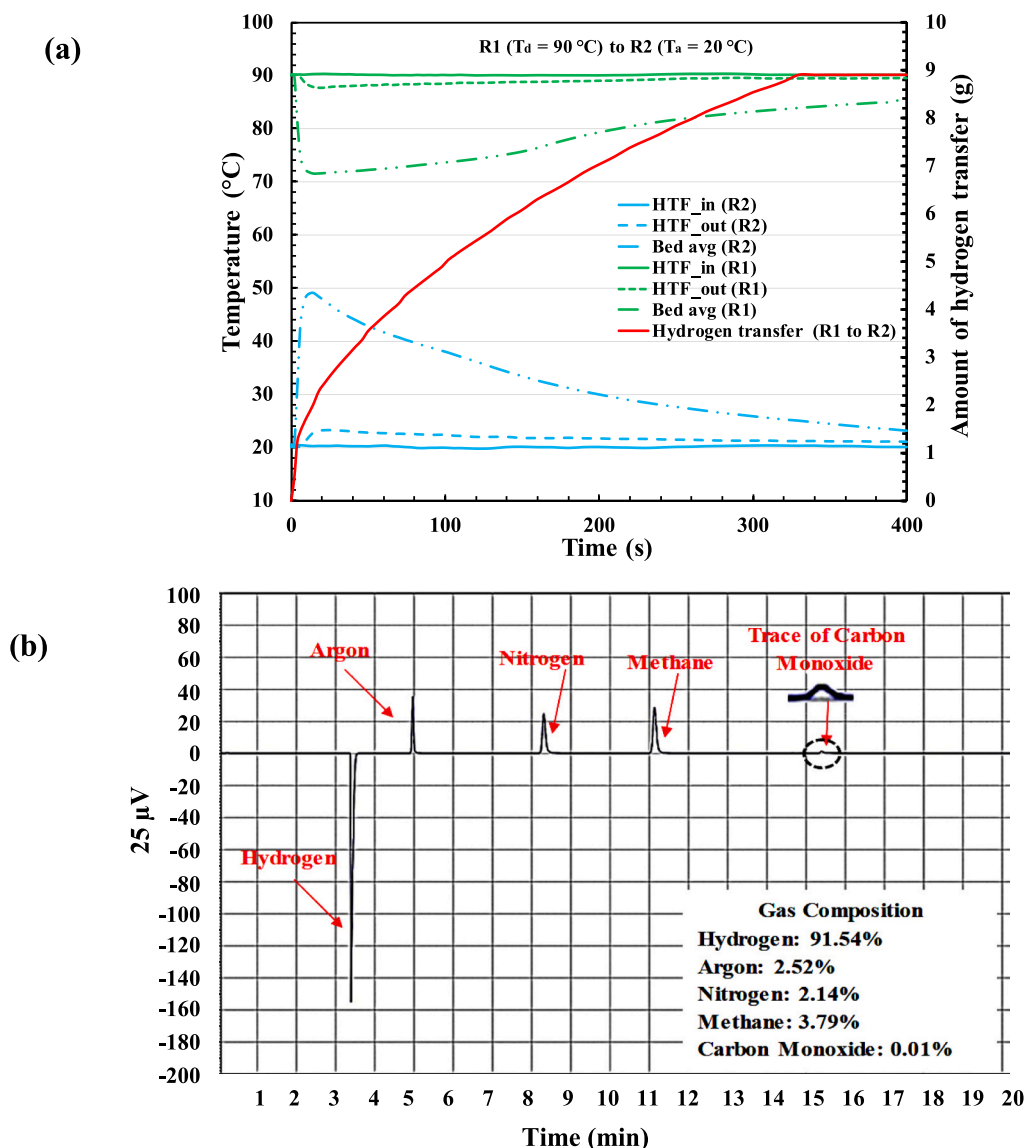


Fig. 9. (a) Gas transfer from stage 1 reactor (R1) to stage 2 reactor (R2) after first stage purification and (b) TCD results of desorption sample after stage 1 purification.

3.2.2. Stage 2 Purification

After completion of stage 1 purification, reactors were prepared for stage 2 purification. Initially, stage 2 reactor was flushed at 15 °C. However, just 0.23 g of gas came out in flushing stage 2, the amount was significantly small to reach gas chromatography, through online sampling like. Therefore, due to unavailability of sufficient gas sample, TCD analysis was not performed for stage-2 flushing. Further, the reactors R2 and R3 were prepared for coupling, wherein R2 and R3 were maintained at 90 °C and 20 °C respectively. Once the respective bed and HTF temperatures stabilized, the reactor valves were opened to facilitate hydrogen transfer from R2 to R3. Amount of hydrogen transferred from R2 to R3 was 8.12 g in 150 s, which constitutes to 91.13% of total hydrogen available for transfer. During this process, the peak bed temperature of 74.8 °C in 18 s and peak HTF temperature of nearly 24.5 °C was attained in the absorbing reactor R3. However, because of endothermic desorption, the bed and HTF temperature of R2 was dropped to 59.6 °C and 86.2 °C respectively (Fig. 10a). Because of lower equilibrium pressure of R3 at 20 °C, the rate of absorption was faster as compared to R1 to R2.

Further, after stage 2 purification, the desorption sample of stage 2 (during gas transfer from R2 to R3) was collected and analysed.

According to the TCD result, the gas composition after stage 2 purification was of 99.996% hydrogen with 0.004% impurity in the form of traces of 0.003% N₂ and 0.001% CH₄ (Fig. 10b).

3.2.3. Stage 3 Purification

Finally, for obtaining ultra-pure hydrogen, stage 3 purification was carried out by desorbing R3 (LaNi_{4.7}Al_{0.3}) at 90 °C. As depicted in Fig. 11a, amount of hydrogen desorbed from R3 was 7.51 g in 210 s, with reversibility of reversibility of 92.4%, wherein the bed and HTF temperature was dropped to 59.3 °C and 82.1 °C respectively. At last, the desorption sample after stage 3 purification (R3) was collected and analysed, wherein ultra-pure hydrogen was obtained with purity level of 99.9999% and negligible or no impurity (Fig. 11b). Because of significant equilibrium pressure of R3 at 90 °C, the high reversibility was due to negligible loss of gas during the flushing process, as the 99.996% pure hydrogen was transferred from R2 to R3.

It was observed that, in each stage of purification, hydrogen recovery was >90%. Considering final recovery (after stage 3 purification) compared to initial absorption (stage 1 absorption), the net hydrogen recovery observed was 78.5%. The main reason for low recovery is the experimentation on small system, wherein there is always 0.5–1 g gas

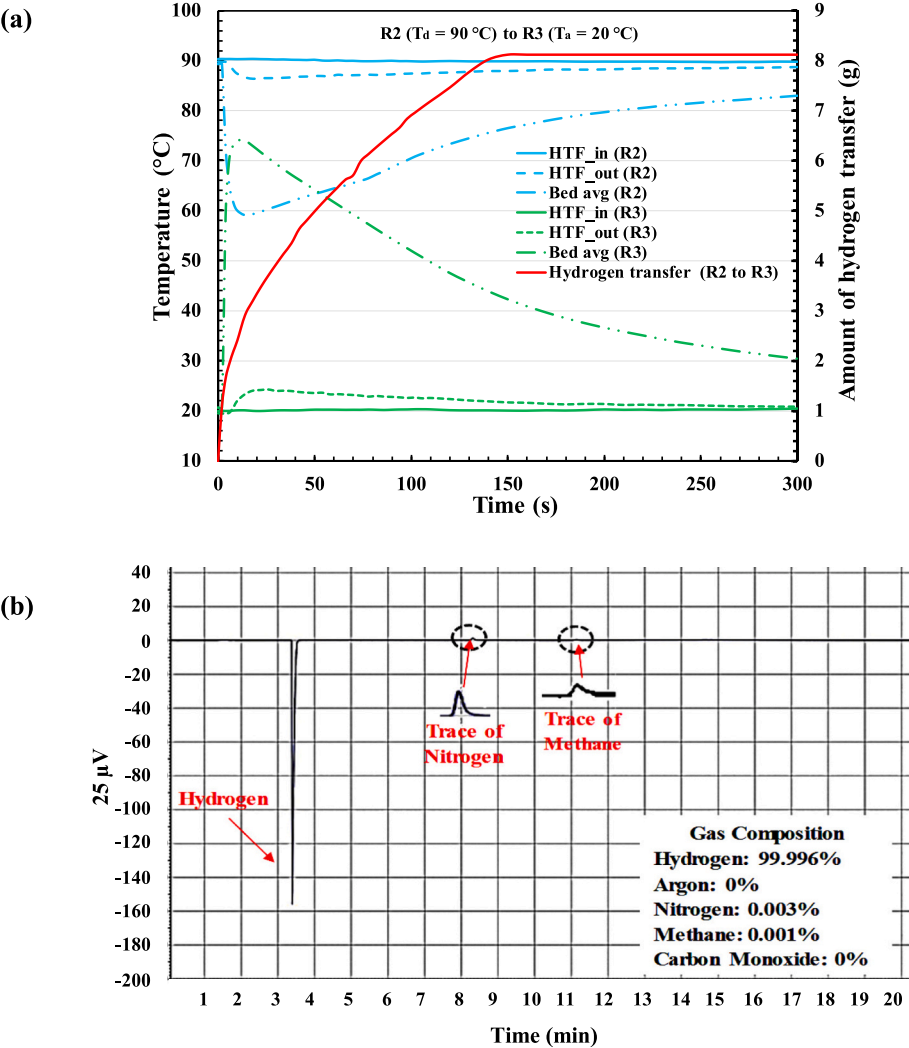


Fig. 10. (a) Gas transfer from stage 2 reactor (R2) to stage 3 reactor (R3) during second stage purification and (b) TCD analysis of desorption sample after stage 2 purification.

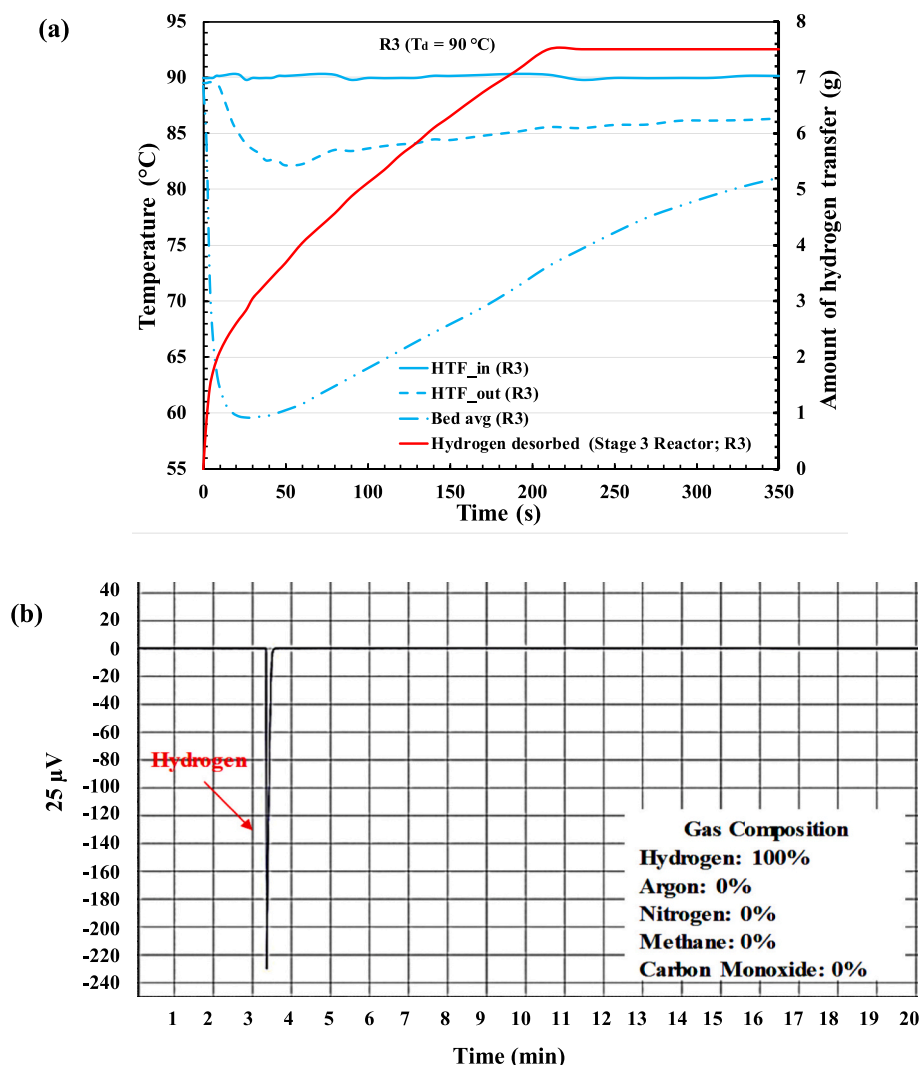


Fig. 11. (a) Plot representing rate of gas transfer and temperature variation during of HTF and MH bed during desorption of pure hydrogen after third stage purification and (b) TCD analysis of desorption sample after stage 3 purification.

Table 5
Summary of gas analysis during multistage hydrogen purification.

Purification stage	Absorbed mass	Flushed mass	Remained mass	Hydrogen transferred	H ₂ Purity	H ₂ recovery
Stage 1 (R1)	10.59 g	1.03 g	9.56 g	8.91 g	91.54%	93.20%
Stage 2 (R2)	8.91 g	0.23 g	8.68 g	8.12 g	99.996%	93.50%
Stage 3 (R3)	8.21 g	Not traceable	8.21 g	7.51 g	99.9999%	91.47%

loss within the gas circuit, due to evacuation and purging. For larger system this amount (0.5–1 g) will not affect much in the calculation of hydrogen recovery. The summary of gas transfer in each stage is depicted in Table 5.

4. Conclusions

The present work reports the hydrogen storage and separation characteristics of metal hydride based multistage hydrogen purification system. The discussed system used different compositions of AB₅ intermetallic alloys to separate hydrogen from an impure gas sample with range of impurity in single or multistage. The multi-reactor system showed hydrogen storage capacity of 1.3–1.4 wt%. Due to alloy tailoring characteristics of the MH alloys, the equilibrium bed pressure can be varied at room temperature, which is in the order La_{0.9}Ce_{0.1}Ni₅ < LaNi₅

< LaNi_{4.7}Al_{0.3}. The variation in equilibrium bed pressure facilitated efficient coupling of the MH reactors for multistage operation. The conclusions drawn from the study are as follows:

- The maximum hydrogen transfer in couple reactor system was observed for temperature slot of 20 °C for absorbing reactor and 90 °C for desorbing reactor, which was in the range of 1.03 wt% to 1.13 wt%. The rate of hydrogen transfer was directly proportional to the pressure difference between the reactors. This pressure difference can be maximised by offering higher temperature difference between the absorbing and desorbing reactor.
- The reactors showed significant hydrogen purification characters, wherein the impurity level of up to 20% can be separated using single stage purification, however with increase in the impurity content the purification can be performed in two stage (20–50% impurity) or 3

stage (50%–90%) impurity. The reversibility of the system can be in the range of 90–99%, however, with further increase in the desorption temperature and decrease in the flushing temperature, the hydrogen losses during different stages of purification can be minimised.

- Regeneration of MH alloys are possible after alloy poisoning, wherein reactivation of alloys can be performed by evacuating MH reactor to 10^{-2} mbar at 90–95 °C, followed by absorption at 20–30 bar and 20–30 °C.
- The absorption, flushing and desorption samples were collected for TCD analysis using helium as carrier gas. The TCD analysis was found to be effective method for detecting gas samples, however, because of maximum impurities flushed during stage 1 purification, the gas volume obtained after flushing of stage 2 and stage 3 reactors was not enough to perform TCD analysis.
- The MH based hydrogen purification technology can be cost effective, as it can be operated using low grade thermal input, and with minimal electrical input which can be drawn from solar. The core of the system, i.e. the reactors, has no moving parts within it. This minimises the maintenance cost of the system as there is no wear and tear.

Author contributions

Alok Kumar: Conceptualization and Visualization, Methodology, System Development, Analysis, Experimental Investigation and Validation, Writing – original draft.

P. Muthukumar: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Validation.

CRediT authorship contribution statement

Alok Kumar: Writing – original draft, Software, Investigation, Data curation, Conceptualization. **P. Muthukumar:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization.

Declaration of competing interest

The system and process disclosed in this work have been filed as Patent (Indian Patent Application No: 202131059468) with Indian Institute of Technology Guwahati as the applicant and P. Muthukumar (Principal Investigator) & Alok Kumar (Research Scholar) at Indian Institute of Technology Guwahati, as inventors. The patent application is granted with patent no: 403389.

Data availability

Data will be made available on request.

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